Name: _____

WORKSHEET

Rates of change

There's only one corner of the universe you can be certain of improving …

7	8	1		14	2	7	14	5		12	11	6	3		

11	10	8		5	4	9	13

Aldous Huxley

Rule

$$\text{Rate of change} = \frac{\text{change in quantity 2}}{\text{corresponding change in quantity 1}}$$

$$\text{Average rate of change} = \frac{\text{second value} - \text{first value}}{\text{time taken for change}}$$

Questions

1 At 12 years of age, Rachel was 142 cm tall. She grew to a height of 163 cm by 18 years of age. What was her average rate of growth?

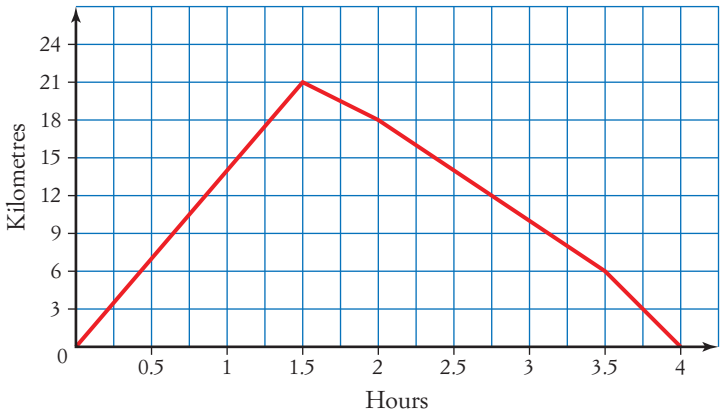
2 A bathtub has 30 L of water in it at a temperature of 18°C. As 25 L of hot water is added, the temperature rises to 23°C. What is the rate of change of temperature with litres added?

3 A grain silo which is filled up to a height of 87 cm has a mass of 298.6 kg. When it is filled to a height of 142 cm, its mass is 452.6 kg. What is the rate of change of mass with height?

4 A courier driver travels from the Gold Coast to Brisbane and back, a round trip of 180 km. On the way to Brisbane he travels at 95 km/h, but on the way back the traffic is heavy and he can only travel at 70 km/h. What is the driver's average speed for the trip?

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A runner's progress over the course of a marathon is recorded in the graph below.



5 What was the runner's slowest speed during the race?

6 What was the runner's speed during the first hour of the race?

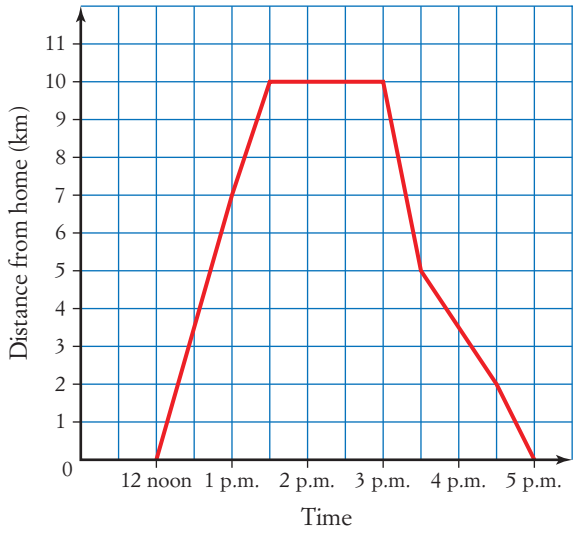
7 What was the runner's speed during the last half-hour of the race?

8 What was the runner's speed from 2 hours to $3\frac{1}{2}$ hours into the race?

9 What was the runner's average speed for the marathon?

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Michael decided to walk his dog over to his friend's house to visit. His journey is shown in the graph below.



10 When was Michael walking the fastest?

11 When was Michael walking the slowest?

12 What was Michael's speed during the first hour of leaving his house to walk to his friend's?

13 What was Michael's speed between 3 p.m. and 3.30 p.m.?

14 What was Michael doing between 1.30 p.m. and 3 p.m.?

Solutions

A 12 km/h

D 3.5 cm/year

E 80.6 km/h

F 10 km/h

H 0.2°C/L

L 10.5 km/h

N 8 km/h

O Between 3.30 p.m. and 4.30 p.m.

R 2.8 kg/cm

S 6 km/h

T Visiting at his friend's house

U 14 km/h

W Between 3 p.m. and 3.30 p.m.

Y 7 km/h

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